

NURIOOTPA, Australia

WMO: 946810

Lat: 34.4761S

Lon: 139.0056E

Elev: 276

StdP: 98.05

Time Zone: 9.50 (AUA)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.5	1.7	-8.1	1.9	26.4	-5.3	2.5	25.5	12.6	11.1	11.5	10.7	1.4	50	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.4	37.5	17.6	35.1	17.0	32.8	16.6	20.9	27.3	19.7	27.8	18.7	27.6	5.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.3	14.6	22.8	17.5	13.0	21.7	15.9	11.6	20.8	61.8	26.3	57.4	27.7	53.9	27.7	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	9.3	8.3	-1.6	41.1	1.1	1.4	-2.4	42.1	-3.1	42.9	-3.7	43.7	-4.5	44.7		
			-1.0	24.8	0.8	3.4	-1.6	27.2	-2.0	29.2	-2.5	31.0	-3.1	33.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	22.0	21.6	18.8	15.5	11.9	9.3	8.9	9.4	11.8	14.4	17.5	19.5
	DBStd	5.85	4.59	4.50	4.14	3.34	2.70	2.24	2.15	2.43	3.00	3.87	4.55	4.58
	HDD10.0	151	0	0	0	1	10	38	47	40	11	3	1	0
	HDD18.3	1649	14	15	43	99	199	271	293	276	196	134	71	39
	CDD10.0	1984	371	324	274	164	70	18	13	22	66	141	226	295
	CDD18.3	441	127	106	59	13	1	0	0	0	2	13	46	76
	CDH23.3	5826	1644	1295	712	181	9	0	0	1	31	240	643	1070
	CDH26.7	2832	882	655	316	53	0	0	0	3	84	289	549	
Wind	WSAvg	4.0	4.3	4.1	3.8	3.4	3.4	3.5	4.0	4.1	4.3	4.3	4.2	4.3
Precipitation	PrecAvg	492	20	19	22	39	54	56	67	63	61	45	30	21
	PrecMax	711	78	100	132	126	173	132	118	117	148	117	73	86
	PrecMin	222	0	0	0	2	14	10	16	15	19	1	0	1
	PrecStd	112	16	24	24	30	36	32	27	30	31	33	19	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	38.7	36.3	31.8	24.6	18.9	18.1	21.7	26.6	32.6	36.4	39.4
		MCWB	18.1	18.8	17.5	15.6	12.9	12.6	10.7	11.9	13.7	15.2	16.5	17.7
	2%	DB	37.9	36.2	32.9	27.7	21.7	16.3	15.8	18.6	23.5	28.9	33.0	35.9
		MCWB	17.6	18.2	16.9	14.2	12.6	10.9	9.9	10.8	12.8	13.9	16.0	16.1
	5%	DB	34.7	33.8	30.3	25.2	19.1	14.8	14.3	16.4	20.7	25.8	30.2	32.5
		MCWB	17.3	17.5	16.3	13.8	12.0	10.6	9.6	10.2	12.3	13.6	15.2	15.6
	10%	DB	31.7	30.9	27.3	22.6	17.2	13.6	13.0	14.5	18.4	22.8	27.0	29.3
		MCWB	16.7	16.9	15.7	13.4	11.6	10.1	9.5	9.7	11.5	12.9	14.5	15.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.3	22.9	20.8	18.3	15.4	13.9	12.6	13.2	15.9	17.7	19.7	23.0
		MCDB	26.7	31.7	27.0	22.3	19.1	16.2	14.6	17.9	21.5	26.4	26.9	25.6
	2%	WB	20.5	20.9	19.2	16.6	14.1	12.3	11.5	12.0	14.4	15.9	18.0	19.4
		MCDB	27.6	28.2	26.6	21.5	17.8	14.3	13.5	16.2	20.7	22.9	26.0	27.2
	5%	WB	19.3	19.6	18.0	15.5	13.2	11.4	10.7	11.1	13.1	14.7	16.9	17.8
		MCDB	29.3	27.4	26.0	21.0	16.8	13.5	12.9	14.7	18.7	21.9	26.1	26.8
	10%	WB	18.3	18.3	16.8	14.4	12.5	10.7	10.0	10.4	12.0	13.6	15.7	16.6
		MCDB	29.0	27.5	25.2	20.5	16.0	12.9	12.1	13.5	17.0	20.6	24.6	25.8
Mean Daily Temperature Range	5% DB	MDBR	14.4	13.6	12.7	11.9	9.5	8.2	8.0	9.2	10.4	12.5	13.5	14.0
		MCDBR	18.3	17.2	16.6	16.4	13.2	10.4	10.2	13.1	15.1	18.0	17.9	18.8
		MCWBR	5.6	5.7	5.5	6.2	6.1	6.1	5.9	6.8	6.9	6.8	6.1	6.0
	5% WB	MCDDBR	13.6	12.8	12.9	11.5	9.4	7.6	7.7	10.2	12.8	14.1	14.7	14.0
		MCWBR	6.4	5.8	5.8	5.8	5.2	5.1	5.1	6.1	7.1	6.6	5.9	6.8
Clear-Sky Solar Irradiance	taub	0.332	0.329	0.313	0.305	0.295	0.284	0.285	0.295	0.320	0.329	0.334	0.326	
	taud	2.509	2.536	2.568	2.576	2.584	2.604	2.593	2.541	2.463	2.452	2.468	2.516	
	Ebn at Noon	1005	987	967	920	875	861	879	916	943	975	995	1015	
	Edh at Noon	113	106	95	84	73	67	71	85	104	114	117	113	
All-Sky Solar Radiation	RadAvg	7.89	6.99	5.72	4.15	2.88	2.37	2.56	3.47	4.74	6.12	7.19	7.74	
	RadStd	0.39	0.34	0.27	0.22	0.13	0.14	0.10	0.24	0.38	0.39	0.34	0.29	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.36	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	-92	+158	+70

Nomenclature: See separate page